

FIFA World Cup 2026

Simulation Forecasts Versus Actual Results

Prepared 8 October 2026 | Four models | 10,000 runs per model

Abstract

This study develops and retrospectively evaluates a probabilistic simulation framework for the FIFA World Cup 2026. The repository combines official squad records, EAFC-style player ratings, public player statistics and national-team strength priors to simulate complete tournaments. Three primary approaches are compared: a Poisson baseline, a heuristic event simulator, and the same event simulator augmented with a trained Random Forest. A legacy Random Forest run is retained as a reference. Each saved forecast contains 10,000 tournament realisations. Predictions are compared with FIFA's published outcomes using stage-wise Brier scores, champion log loss and top- k qualifier overlap. Spain, the eventual champion, received title probabilities of 10.26%, 18.18% and 17.11% in the primary models, although none ranked Spain first. Heuristic v2 achieved the lowest mean six-stage Brier score, 0.0840, compared with 0.0872 for the baseline and 0.0867 for Random Forest v2. Both v2 approaches ranked Kylian Mbappé first by expected goals. The analysis also identifies substantial errors in Germany's and Brazil's projected advancement and in the prospects of Norway and Morocco. Interpretation is limited by heterogeneous player-data coverage, synthetic model training, and the absence of independently audited pre-tournament publication timestamps. The results support the value of a transparent simulation pipeline while motivating historical backtesting, probability calibration and better source-version tracking.

Keywords: Monte Carlo simulation; football forecasting; Poisson model; Random Forest; Brier score; data provenance.

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1 Introduction and Project Objectives

Tournament forecasting involves several sources of uncertainty: which players are available, how team strength translates into match outcomes, and how the draw and knockout path shape progression. A team can have the greatest chance of winning while still being more likely to lose the tournament than to win it. The project’s central output is therefore a distribution of possible outcomes.

The repository began with an interpretable baseline and subsequently added player-performance information, team-rating priors and a more detailed match engine. Keeping the baseline makes it possible to ask whether additional complexity improves the forecasts, and where it fails to do so.

1.1 Research Questions

1. How much forecast probability did each model assign to the eventual champion?
2. How accurately did the models allocate advancement probabilities across all 48 teams?
3. Did the event simulator and learned outcome layer improve on the simpler baseline?
4. Which unexpected results reveal weaknesses in team strength, tournament routing or data coverage?
5. Which conclusions are supported by saved evidence, and which require further validation?

1.2 Scope and Contributions

The project covers a 48-team competition with 12 groups of four, followed by a 32-team knockout phase. The top two teams in each group and eight best third-place teams progress. Outputs include title and stage probabilities, player goal and assist summaries, match-event totals, model explanations and data-quality reports.

This report brings together the data pipeline, model definitions and evaluation of the saved forecasts. The comparison uses stored results without retraining on the observed tournament. The current source code documents the implemented mechanisms, but its presence alone does not establish an immutable code version for every June run. Frozen JSON metadata are the primary evidence for each run’s inputs, seed and model mode.

1.3 Observed Tournament Outcome

Spain defeated Argentina 1–0 after extra time in the final. England finished third and France fourth. Mbappé won the Golden Boot with 10 goals [15, 13, 14]. These observations supply evaluation targets; they are not used to update the frozen forecasts.

2 Datasets, Sources and Preparation

The project uses several datasets with different coverage, units and time granularities. A player appearing in the combined file does not imply that every player has a complete, comparable recent-season record.

2.1 Official Squads

The working roster contains 48 teams and 1,248 players. Its metadata identify FIFA’s English squad-list PDF, Version 1, dated 3 June 2026 at 11:30 UTC, as the source [1]. The script `rebuild_squads_from_fifa_pdf.py` extracts squad records; the active output is `guardian_world_cup_2026_player_guide.json`. Despite its inherited filename, the current file identifies FIFA as its squad source. The Guardian guide is separately listed in tournament-structure provenance [11].

The source URL is a live resource: when checked for this report, the online PDF showed a later July timestamp. That page cannot, by itself, prove the exact contents of the locally recorded June snapshot. Archive the original file and its checksum when reproducing or extending the study.

2.2 EAFC Player Ratings

The local `eaafc26_players.csv` contains 18,405 records. Its update-date column is 19 September 2025 throughout the file. It supplies player names, nationality, positions, overall ratings and additional attributes. The saved baseline reports 980 matched players and 268 fallback ratings out of 1,248: approximately 78.5% matched coverage and 21.5% fallback coverage.

The export includes SoFIFA-hosted image URLs and player-path fields. These are provenance clues, not proof of the file’s original distributor. The repository does not preserve a verified download URL, dataset release identifier or licence for this exact CSV. Accordingly, this report cites the local artifact and records that upstream provenance gap explicitly [2].

2.3 Public Player-Performance Data

The v2 input is `player_performance_data_statbunker.json`. Its metadata record 19,923 source rows and 1,248 output players. Source rows can repeat a player across competitions, seasons and table types; they are not 19,923 unique players. StatBunker supplies the main public competition-table input, including appearances, goals, assists, cards and goalkeeper clean sheets [3].

Transfermarkt-derived CSVs from `salimt/football-datasets` supplement player identity and performance information [4]. English Wikipedia biographies and manual public-web research fill some remaining gaps [5]. Two players have SofaScore-derived aggregate statistics recorded in the final metadata [6]. The final source-summary counts are shown in Table 1.

Table 1. Primary player-statistics sources in the combined input.

Primary source	Players	Share (%)
Manual Web Research	31	2.5
Sofascore	2	0.2
Statbunker	723	57.9
Transfermarkt	283	22.7
Wikipedia	209	16.7

2.4 Coverage and Missingness

StatBunker’s default pages do not supply complete minutes, expected goals, expected assists, progressive passing, tackles, saves or match ratings. Wikipedia infoboxes often contain career totals rather than recent-season splits. Transfermarkt supplements do not make all records equivalent in detail or freshness. The metadata’s “complete” flag describes populated player records; two partial-context rows remain documented.

The research coverage report illustrates the unevenness. Panama has 18 players below the stored 0.55 confidence threshold and 24 with zero recorded minutes; Senegal has 14 and 25 respectively. A zero in these source-derived fields can reflect missing evidence rather than a player having played no football. The stored confidence measure is an engineering heuristic, not a calibrated probability that the record is correct.

An earlier squad-quality report recorded 17 squad-shape issues and a `needs_refresh` status. It predates later enrichment, so it should be treated as a historical diagnostic requiring a fresh check, not proof that all 17 issues remain in the current roster.

2.5 National-Team Rating Priors

The primary v2 forecasts used FIFA ranking positions transformed to an Elo-like scale [7]:

$$R_{\text{prior}} = 2170 - 130 \ln(r), \quad (1)$$

where r is ranking position. The preserved fallback file records a ranking date of 19 November 2025. This is a model-defined transformation of ranking positions, not FIFA ranking points and not a historical World Football Elo series.

A subsequent import brought in 177 World Football Elo team records with an as-of date of 11 June 2026 [8]. The import report shows coverage of all required teams. These newer inputs are present locally, but the frozen v2 runs evaluated here still declare the older rank-derived prior. The current input file and the input version used by a saved result must therefore be distinguished.

2.6 Tournament and Venue Context

Groups and tournament rules are stored in `world_cup_2026_simulation_structure.json`, whose metadata cite FIFA draw information and the Guardian player guide [10, 11]. `world_cup_2026_venue_context.json` provides group-region and stage-level context. Heat, humidity and travel-load values are modelling priors; they are not measured match-day weather observations or a complete travel itinerary.

2.7 Identity Resolution and Enrichment

The preparation pipeline has four steps:

1. Extract squad identities, broad positions and clubs from the official list.
2. Normalize names and compare candidates using nationality, name similarity, exact aliases and club information. Review overrides are stored separately.
3. Attach ratings and detailed positions. For unmatched EAFC players, use role-based defaults: 68 for goalkeepers/defenders and 69 for midfielders/forwards.
4. Merge public statistics with source identity, collection status and confidence, then construct team profiles from player records.

Name normalization supports matching across sources; presentation should retain the player’s proper spelling. The report therefore uses Kylian Mbappé. Some archived records retain unaccented matching keys; those frozen identifiers are preserved to keep the evaluation reproducible.

`performance_data_config.json` specifies intended collection windows, competition weights and stage multipliers. These settings describe collection and modelling intentions. A listed field or provider is not evidence that it was available for every player or consumed by every saved simulation.

2.8 Attempted and Future Data Sources

SofaScore access was explored through direct and browser-assisted collectors; the final source summary records only two primary SofaScore rows. Sportmonks, API-Football/API-Sports and FootyStats have provider-ready collection paths, but the saved input summary does not identify them as primary suppliers.

The project also attempted to fetch the international-results dataset maintained by `martj42/international_results` [9]. The June manifest records a failed download and no normalized results file. The stored Random Forest training metadata consequently report zero historical rows. This dataset is a planned input for supervised backtesting, not training evidence for the forecasts evaluated here.

2.9 Dataset Register

The accompanying `dataset_sources.csv` records each source, the corresponding local artifact, how it was used, and its provenance limitations. Provider URLs are distinguished from verified URLs for the exact snapshot. The register should be updated whenever inputs change.

3 Models and Simulation Architecture

3.1 Monte Carlo Tournament Simulation

Monte Carlo simulation repeatedly draws match outcomes and plays through the competition. For event A , its estimated probability after N tournaments is

$$\hat{P}(A) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{A \text{ occurs in run } i\}. \quad (2)$$

Here $N = 10,000$ for each compared artifact. This is a count of simulated tournaments, not machine-learning training epochs. Under independent runs, the conditional Monte Carlo standard error is approximately $\sqrt{\hat{p}(1-\hat{p})/N}$. For a 20% title probability it is about 0.4 percentage points. Running more simulations reduces this sampling noise; it does not repair biased ratings or inaccurate football assumptions.

3.2 Model 1: EAFC–Poisson Baseline

The baseline in `simulate_world_cup.py` builds a position-aware starting XI, a top-18 group and a full-squad mean. Team strength is

$$S = 0.62\bar{R}_{\text{XI}} + 0.28\bar{R}_{18} + 0.10\bar{R}_{\text{squad}}. \quad (3)$$

For strength difference $\Delta = S_A - S_B$, expected goals are

$$\lambda_A = \text{clip}(1.28 + 0.045\Delta, 0.25, 3.4), \quad (4)$$

$$\lambda_B = \text{clip}(1.28 - 0.045\Delta, 0.25, 3.4). \quad (5)$$

The match score is sampled using Poisson goal counts. The baseline is useful because its rating-to-score relationship is simple to inspect. Its limitations include static strength, role-based fallback ratings and limited representation of changing match state. Its stored metadata describe knockout seeding based on group-stage performance, whereas v2 uses constrained bracket routing; differences between model results therefore reflect more than the learning algorithm alone.

3.3 Shared v2 Event Engine

The v2 implementation adds position-sensitive player-performance adjustments, lineup selection and separate attack, midfield, defence, goalkeeper and possession profiles. Match context includes team priors, host advantage, rest, accumulated fatigue, lineup replacements and venue/travel assumptions. Regulation time is compressed into nine simulated ticks and extra time into three. Possession, fouls, cards, removals, substitutions, set pieces and game state can affect scoring; tied knockout matches proceed to extra time and penalties. The engine carries player state through the tournament and supports CPU-parallel tournament batches.

3.4 Model 2: Heuristic Event Simulator

The heuristic mode uses 96 deterministic, seeded rule-based votes to adjust attack share, tempo, discipline and possession. The thresholds are hand-built mechanics, not trees learned from observed match outcomes. The aggregate modifiers feed the expected-goal calculation before the event engine produces the match result.

This version asks whether additional team and match context improves on the baseline without fitting a supervised outcome classifier. It is more expressive, but the additional coefficients and constraints also introduce assumptions requiring validation. Both v2 modes retain this heuristic component.

3.5 Model 3: Random Forest-Augmented Event Simulator

The learned component is a scikit-learn Random Forest classifier with outcomes A win, draw and B win. The inspected implementation uses 96 trees, maximum depth 8, at least 6 samples per leaf, balanced-subsample class weighting and a fixed seed. Its 21 inputs cover strength and unit differences, discipline and data confidence, host/rest/fatigue context, venue/travel conditions and tournament stage information. A model feature need not represent directly measured evidence: several inputs are derived priors.

The forest modifies the event simulator rather than replacing it. Let s_{base} be the engine’s attacking share and p_A, p_D, p_B the forest’s outcome probabilities. The pre-clipping blend is

$$s_A = 0.72s_{\text{base}} + 0.28 \frac{p_A}{\max(0.01, p_A + p_B)}. \quad (6)$$

The draw probability also adjusts total expected goals. Consequently, the final tournament probabilities reflect the combined rating model, heuristic rules, learned adjustment and stochastic event engine.

3.6 Synthetic Training and Its Implications

The stored forest uses 6,768 synthetic rows and no historical match rows. This count follows from 48×47 ordered team pairings across three contexts. The generator perturbs features and samples outcome labels from a hand-defined probability function involving team strength, unit matchups and context.

The model therefore learns the behaviour of the synthetic generator. Good performance on held-out synthetic rows does not establish real-match accuracy. The saved RF run’s 0.7301 accuracy is measured on its training data and must not be presented as a 73.01% World Cup prediction success rate. The code supports historical training when enough usable labels exist; that path was not active for these artifacts.

3.7 Legacy Reference and Experimental Controls

The legacy RF run uses squad-proxy Elo rather than the national-team rank prior. It is retained to expose the effect of earlier assumptions, but excluded from the primary model-average comparison. Different seeds, bracket mechanics, priors and context versions mean the stored runs are not a controlled ablation of one factor at a time.

Table 2. Frozen forecast inventory. Seeds are identifiers, not dates of publication.

Model	Runs	Seed	Rating prior
EAFC baseline	10000	20260605	EAFC only
Heuristic v2	10000	20260620	FIFA rank-derived
Random Forest v2	10000	20260621	FIFA rank-derived
Legacy RF	10000	20260618	Squad proxy

3.8 Additional Research Models

The research pipeline also benchmarks L2 logistic regression, sigmoid-calibrated logistic regression, Extra Trees, histogram gradient boosting and an ensemble weighted by inverse validation log loss. Logistic regression provides a linear reference; calibration adjusts probability outputs; the tree ensembles explore nonlinear feature interactions. These are diagnostic experiments, not five additional frozen tournament forecasts in this retrospective comparison.

The stored research split contains 5,076 training rows and 1,692 test rows from the synthetic frame. Calibrated logistic regression has log loss 0.75862; Random Forest has 0.79449. These numbers use synthetic targets and are not comparable to champion log loss on the actual tournament.

The ensemble chooses its weights using the same held-out results on which the ensemble is then scored. Its reported score is consequently not an independent final test. Future work should separate fitting, calibration, model selection and final chronological evaluation. Random splitting of related synthetic team pairings is also weaker evidence than held-out tournaments or dates.

3.9 Feature Importance and Explainability

The earlier feature study uses tree importance, permutation importance, feature-group ablation, partial dependence, individual conditional expectation and local feature occlusion. Squad-rating difference, possession difference and attack-versus-defence are leading signals in the stored permutation results. Unit-matchup ablation yields the largest reported validation log-loss increase, approximately 0.0228.

These results explain sensitivity within the synthetic modelling framework; they do not establish causal determinants of real football outcomes. The tournament-level RF ablation uses only 30 runs per scenario and leaves the hand-built expected-goal mechanics active. Those exploratory results are not equivalent to full-engine ablation or a 10,000-run controlled comparison.

4 Evaluation Design

The comparison joins frozen forecast percentages with binary indicators of actual advancement for each of the 48 teams. The analysis verifies that the comparison CSVs match the original JSON outputs and that the observed stage memberships are nested correctly. FIFA’s knockout results supply advancement membership [12]. The main comparison contains 1,536 team-stage rows: four models, 48 teams and eight reported outcomes per team.

How the comparison was made

Saved forecasts are evaluated without retraining.



Outcome checks: 32 -> 16 -> 8 -> 4 -> 2 -> 1; CSV forecasts checked against original JSON.

Figure 1. Workflow for evaluating saved forecasts against sourced tournament outcomes.

4.1 Probability Scores

For team t at stage s , let $p_{t,s}$ be the forecast probability and $y_{t,s} \in \{0,1\}$ indicate whether that team reached the stage.

$$BS_s = \frac{1}{48} \sum_{t=1}^{48} (p_{t,s} - y_{t,s})^2, \quad (7)$$

$$Skill_s = 1 - \frac{BS_s}{q_s(1 - q_s)}, \quad q_s = \frac{k_s}{48}, \quad (8)$$

$$LL_{\text{champion}} = -\ln p_{\text{Spain, champion}}. \quad (9)$$

Lower Brier score and log loss are better. Positive skill indicates improvement over a uniform stage probability. The categorical champion Brier score is the *sum* across 48 outcomes; advancement Brier scores are *means*.

The combined advancement score is the equally weighted mean over six stages: last 32, last 16, quarter-finals, semi-finals, final and champion. Third- and fourth-place outcomes are reported separately.

4.2 Top- k Overlap

For a stage admitting k teams, the analysis counts how many of the k highest forecast teams actually reached it. Equal probabilities use alphabetical ordering. This ranking measure complements probability scores but does not assess calibration.

5 Championship Forecasts

Spain had substantial probability in each primary forecast, but no primary model selected Spain as its favourite. A likely contender winning is compatible with the forecast even when the highest-ranked team's prediction is incorrect.

Who each model expected to win

Title probabilities (%): Spain is the actual champion. Identical colours identify the same team.

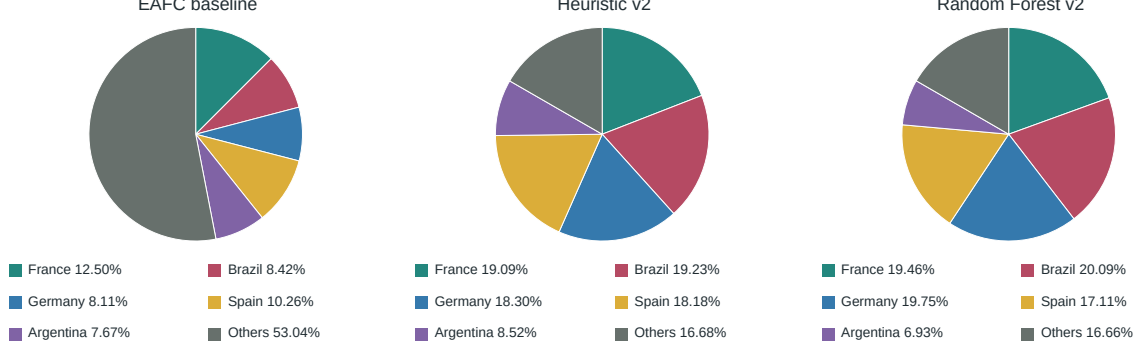


Figure 2. Title-probability distributions for the three primary models. Each pie totals 100%; Others combines all teams outside the five named contenders.

Table 3. Forecast support for the actual champion. Legacy RF is a reference model.

Model	Favourite	Spain rank	Spain (%)	Log loss	Brier sum
EAFC baseline	France	2	10.26	2.277	0.864
Heuristic v2	Brazil	4	18.18	1.705	0.790
Random Forest v2	Brazil	4	17.11	1.766	0.817
Legacy RF	France	2	14.38	1.939	0.870

Spain's forecast title probability

Spain won; probabilities describe the forecast before the observed outcome.

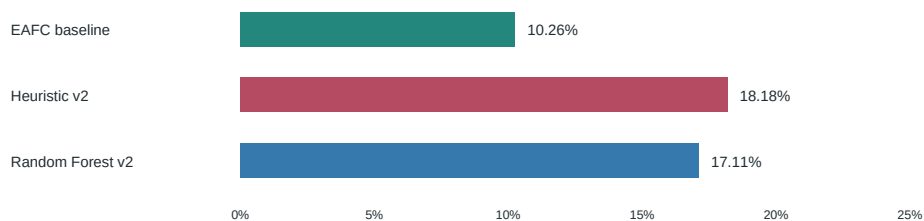


Figure 3. Spain's title probability across the primary models.

6 Advancement Accuracy

The richer heuristic model improves the combined score relative to the baseline, but the improvement is not uniform across stages. The baseline has lower Brier error at the quarter-final, semi-final and final stages.

Forecast error across tournament stages

Mean Brier score across 48 teams; lower is better. Six stages receive equal weight.

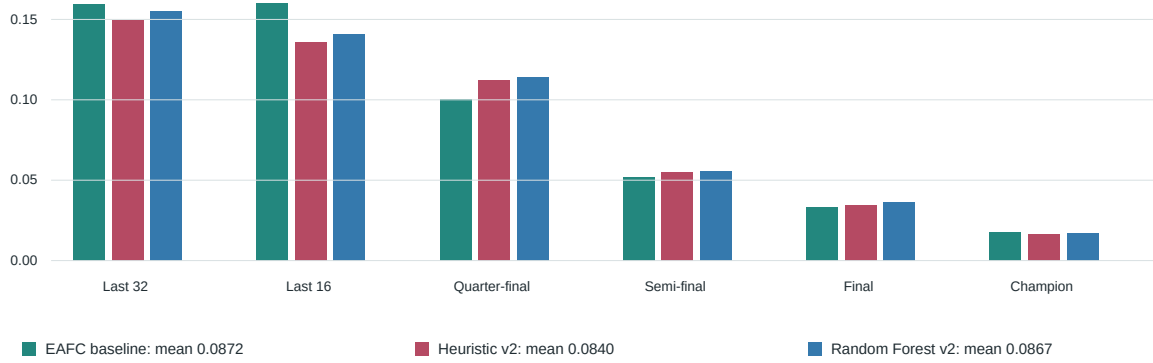


Figure 4. Stage-wise mean Brier scores. Compare models within a stage because the stage base rates differ.

Table 4. Advancement error and top-ranked qualifier overlap across the three primary models.

Stage	Baseline	Heuristic	RF	Top- <i>k</i> hits (B / H / RF)
Last 32	0.1594	0.1497	0.1551	27/32 / 27/32 / 27/32
Last 16	0.1599	0.1357	0.1411	11/16 / 11/16 / 11/16
Quarter-final	0.1004	0.1124	0.1143	4/8 / 4/8 / 4/8
Semi-final	0.0521	0.0552	0.0559	2/4 / 2/4 / 2/4
Final	0.0332	0.0345	0.0366	1/2 / 0/2 / 0/2
Champion	0.0180	0.0165	0.0170	0/1 / 0/1 / 0/1
Third place	0.0190	0.0187	0.0174	0/1 / 0/1 / 0/1
Fourth place	0.0197	0.0211	0.0211	0/1 / 0/1 / 0/1

Table 5. Equal-weight combined advancement scores; lower is better.

Primary model	Mean six-stage Brier
EAFC baseline	0.0872
Heuristic v2	0.0840
Random Forest v2	0.0867

7 Where Forecasts and Outcomes Diverged

Forecasts versus actual advancement

Random Forest v2; outlined cells indicate teams that actually reached the stage.

	Last 32	Last 16	Quarter-final	Semi-final	Final	Champion
Spain	99.8%	88.5%	53.0%	39.2%	26.6%	17.1%
Argentina	99.9%	83.6%	57.5%	37.0%	14.2%	6.9%
England	99.9%	92.7%	75.4%	34.5%	13.5%	6.8%
France	99.8%	95.5%	52.1%	40.0%	29.4%	19.5%
Brazil	99.8%	86.8%	78.0%	68.8%	42.1%	20.1%
Germany	99.8%	93.1%	82.2%	72.4%	42.9%	19.8%
Norway	89.9%	48.0%	11.4%	3.5%	0.8%	0.1%
Morocco	91.0%	27.6%	12.6%	4.2%	0.8%	0.1%
Belgium	99.9%	90.2%	39.9%	10.2%	3.7%	1.1%
Switzerland	98.2%	69.2%	20.9%	8.2%	1.8%	0.2%

Darker fill = higher forecast probability. Outline = actual advancement. This is a selected-team view.

Figure 5. Random Forest v2 probabilities for selected teams. Outlined cells indicate actual advancement; darker fills indicate higher forecast probability.

Germany exited in the last 32 and Brazil in the last 16 despite high v2 advancement probabilities. Norway, Morocco, Belgium and Switzerland reached the quarter-finals. The selected-team heatmap illustrates both overestimation of highly rated contenders and underestimation of unexpected progress.

These discrepancies suggest reviewing upset frequency, bracket effects and player-data coverage. The observed outcomes alone cannot identify which mechanism caused a forecast error.

Table 6. Eight largest absolute discrepancies among primary-model quarter-final, semi-final and final forecasts.

Model	Team	Stage	Forecast (%)	Reached?
Heuristic v2	Norway	Quarter-final	10.38	Yes
Random Forest v2	Norway	Quarter-final	11.41	Yes
Random Forest v2	Morocco	Quarter-final	12.61	Yes
EAFC baseline	Argentina	Final	13.86	Yes
Random Forest v2	Argentina	Final	14.16	Yes
Heuristic v2	Morocco	Quarter-final	15.66	Yes
Heuristic v2	Argentina	Final	17.25	Yes
EAFC baseline	Spain	Final	17.77	Yes

8 Player Scoring and Interpretation

Kylian Mbappé won the Golden Boot with 10 goals. Both primary v2 models ranked him first by expected aggregate goals; the baseline ranked him second.

Table 7. Expected scoring rank and mean goals versus one observed tournament.

Model	Mbappé rank	Mean goals	Actual goals
EAFC baseline	2	1.446	10
Heuristic v2	1	2.946	10
Random Forest v2	1	2.904	10
Legacy RF	1	3.467	10

Expected goals over many simulated tournaments and observed goals in a single tournament are different quantities. A mean of approximately three goals does not predict that the player must score three. Similarly, a ranking by expected goals is not the probability of winning the Golden Boot. This comparison is descriptive; award forecasting would require a separate probability calculation.

9 Provenance, Limitations and Next Steps

9.1 Forecast Provenance

Each model ran 10,000 tournaments. The two primary v2 artifacts used FIFA-rank-derived Elo-scale priors, rather than the raw World Football Elo data imported later. Random Forest training used synthetic profile calibration; historical-match predictive accuracy was not established.

Seeds that resemble dates are random-number seeds, not creation timestamps. The artifacts were saved around the start of the tournament and the v2 files were last modified on 11 June. Local timestamps do not prove publication before kickoff. This report therefore evaluates saved forecasts retrospectively.

9.2 Statistical and Data Limits

Advancement indicators are dependent; six stages across 48 teams are not 288 independent observations. One tournament cannot establish calibration or model superiority. Wilson intervals in prior reports measure Monte Carlo sampling error conditional on model assumptions, not uncertainty about those assumptions.

Saved outputs do not supply probabilities for every actual match or every player. Match-level log loss and full player-goal errors are outside scope. Group-stage exits are inferred from absence in the verified last-32 field. Rankings among teams eliminated at the same stage are not inferred.

9.3 Recommended Research Work

1. Backtest several historical tournaments using information available at each match date.
2. Calibrate goal, draw and upset rates on real international results.
3. Review bracket routing and concentration of probability on stronger teams.
4. Improve low-confidence player coverage and model Golden Boot probabilities separately.
5. Publish dated forecast snapshots before future tournaments.

9.4 Reproducibility

Run `python compare_simulation_with_actuals.py` from the repository root. The analysis regenerates CSV score tables, Markdown and HTML reports, charts and this LaTeX source. Compile the source from the report folder using `pdflatex -jobname=comprehensive_report`

`comparison_report.tex`

Repeat until contents pages and cross-references settle.

Supporting data: `actual_outcomes.json`, `model_scores.csv`,

`team_stage_comparison.csv`, `champion_comparison.csv`,

`scorer_comparison.csv`, `prediction_sha256.json`.

The hashes identify the frozen inputs. Figure PDFs are supplied in the `figures/` folder.

10 Conclusion

The project establishes an inspectable chain from data collection to simulated tournaments and retrospective evaluation. Spain was a credible contender in every primary forecast, but the favoured team was wrong in each case. Heuristic v2 achieved the best combined advancement score on this tournament, while the baseline performed better at some later stages. The RF extension did not deliver a consistent improvement over the heuristic engine.

The most useful next step is stronger validation: chronologically separated historical data, measured match context, controlled model comparisons and published forecast snapshots. These improvements would make it possible to distinguish the benefits of richer football information from the effects of hand-set assumptions and synthetic training.

References and Data Sources

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A Repository and Reproduction Guide

A.1 Pipeline Map

Squad preparation `rebuild_squads_from_fifa_pdf.py` and `enrich_squads_with_eafc.py` prepare the roster and attach ratings.

Performance enrichment

`collect_player_performance_data_statbunker.py`, `integrate_transfermarkt_data.py` and `collect_wikipedia_player_stats.py` collect and merge player evidence.

Tournament simulation `simulate_world_cup.py` implements the baseline; `simulate_world_cup_v2.py` implements the event engine and optional RF layer.

Research diagnostics `analyze_feature_importance.py` and `run_research_upgrade_pipeline.py` generate sensitivity and calibration studies.

Retrospective evaluation `compare_simulation_with_actuals.py` checks the saved probabilities, scores outcomes and regenerates report inputs.

Typesetting `comparison_latex.py` combines checked tables and the `report_templates/` text. `comparison_visuals.py` creates vector figures.

A.2 Reproduction Procedure

1. Retain the four frozen result JSONs, the model-comparison CSVs, actual outcome file and dataset register. Check forecast hashes before evaluating.
2. Run `python compare_simulation_with_actuals.py` at repository root. This regenerates metrics, chart assets and the complete LaTeX source.
3. Compile `comparison_report.tex` from its report directory.
Use job name `comprehensive_report` and keep the adjacent `figures/` directory. Additional passes may be needed until contents pages and citations settle.
4. Review the PDF for table and figure placement. Package the compiled PDF, LaTeX source and vector figures together for sharing.

The simulation source inputs are partly excluded from Git. Reproducing this evaluation from frozen outputs is different from rerunning the full original data-collection and simulation pipeline. The latter requires the original source files, compatible dependencies and archived input versions. Pin software versions and record the code commit when publishing future forecasts.

A.3 Terminology

Prior A pre-existing strength estimate or assumption used by a model.

Calibration Agreement between forecast probabilities and observed frequencies across repeated cases.

Brier score Squared probability error; lower values indicate better forecasts.

Monte Carlo error Sampling noise caused by a finite number of simulated tournaments.

Ablation An experiment removing or neutralizing a feature or mechanism to study sensitivity.

Fallback rating A default value substituted when no sufficiently matched rating is available.